SWITCHING AND VLANS

Course: Diploma in Cyber Security
Topic: Switching and VLANs
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Introduction

A switch connects devices inside a local network. VLANs can divide one physical switch network into separate logical networks. In this assignment, I have covered a simple VLAN diagram and basic Cisco Packet Tracer commands for creating and changing VLANs

Task 1: Devices and VLAN Diagram

Example setup: Laptop = VLAN 10, Friend's Phone = VLAN 20, Smart TV = VLAN 10.

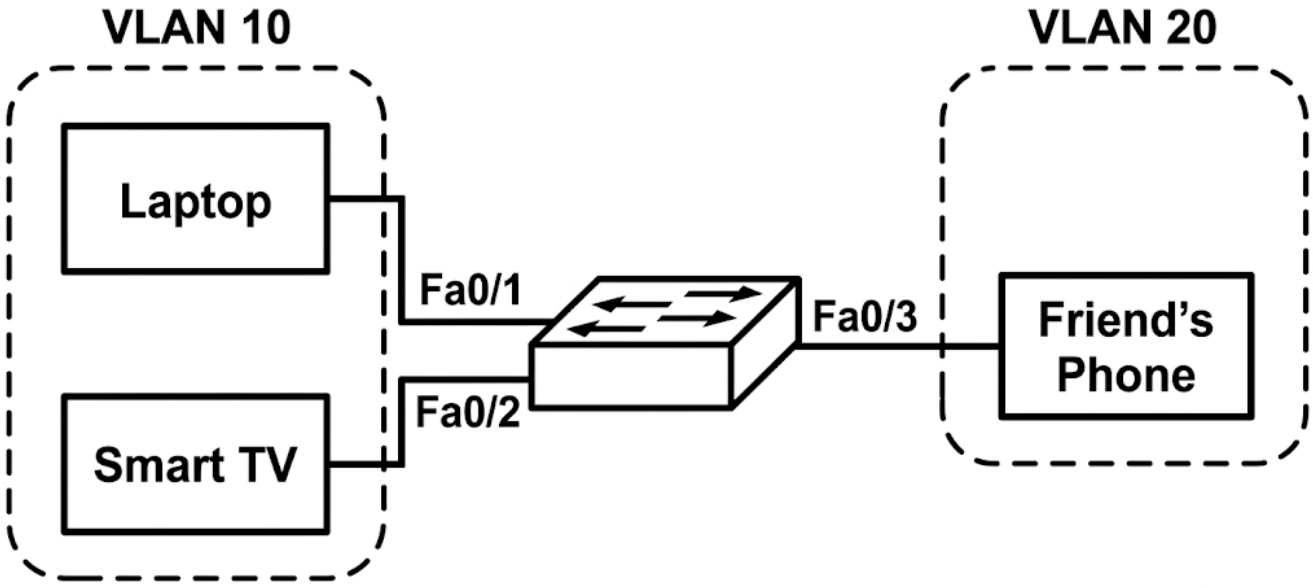


Figure 1: Simple Switch Network with VLAN 10 and VLAN 20

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Task 2: Create and Test Two VLANs

Example Cisco Packet Tracer commands:

```
enable
configure terminal
vlan 10
name STREAMING
vlan 20
name PERSONAL
```

```
interface range fa0/1-3
switchport mode access
switchport access vlan 10
interface range fa0/4-5
switchport mode access
switchport access vlan 20
end
```

Connect three PCs to Fa0/1-Fa0/3 and two PCs to Fa0/4-Fa0/5. Give the PCs suitable IP addresses. PCs in different VLANs will not normally ping each other without inter-VLAN routing.

PRACTICAL: Do this in Packet Tracer/NetSim. Take a screenshot of the VLAN configuration and the failed ping between different VLANs if your teacher requires proof.

Task 3: Move a PC from VLAN 10 to VLAN 20

Before:

```
interface fa0/1
switchport mode access
switchport access vlan 10
```

After:

```
interface fa0/1
switchport mode access
switchport access vlan 20
```

After changing the VLAN, the PC becomes part of VLAN 20. Communication with VLAN 10 needs inter-VLAN routing.

PRACTICAL: In Packet Tracer, change the port and test with ping. A screenshot of the before/after configuration and ping test is useful.

Task 4: Assign 10 Ports to VLAN 30

Example commands:

```
enable
configure terminal
vlan 30
name GAMING
interface range fa0/1-10
switchport mode access
switchport access vlan 30
end
```

This assigns FastEthernet ports 1 through 10 to VLAN 30.

PRACTICAL: Test in Packet Tracer if available. Use 'show vlan brief' to verify the ports are listed under VLAN 30.

Conclusion

In this assignment, I learned how a switch connects devices and how VLANs separate devices into different logical groups. I also learned basic commands to create VLANs, assign ports and move a device from one VLAN to another.

Quick Revision

- VLAN = logical separation of a network.
- Access port = normally carries traffic for one VLAN.
- switchport access vlan = assigns an access port to a VLAN.
- show vlan brief = checks VLAN and port membership.